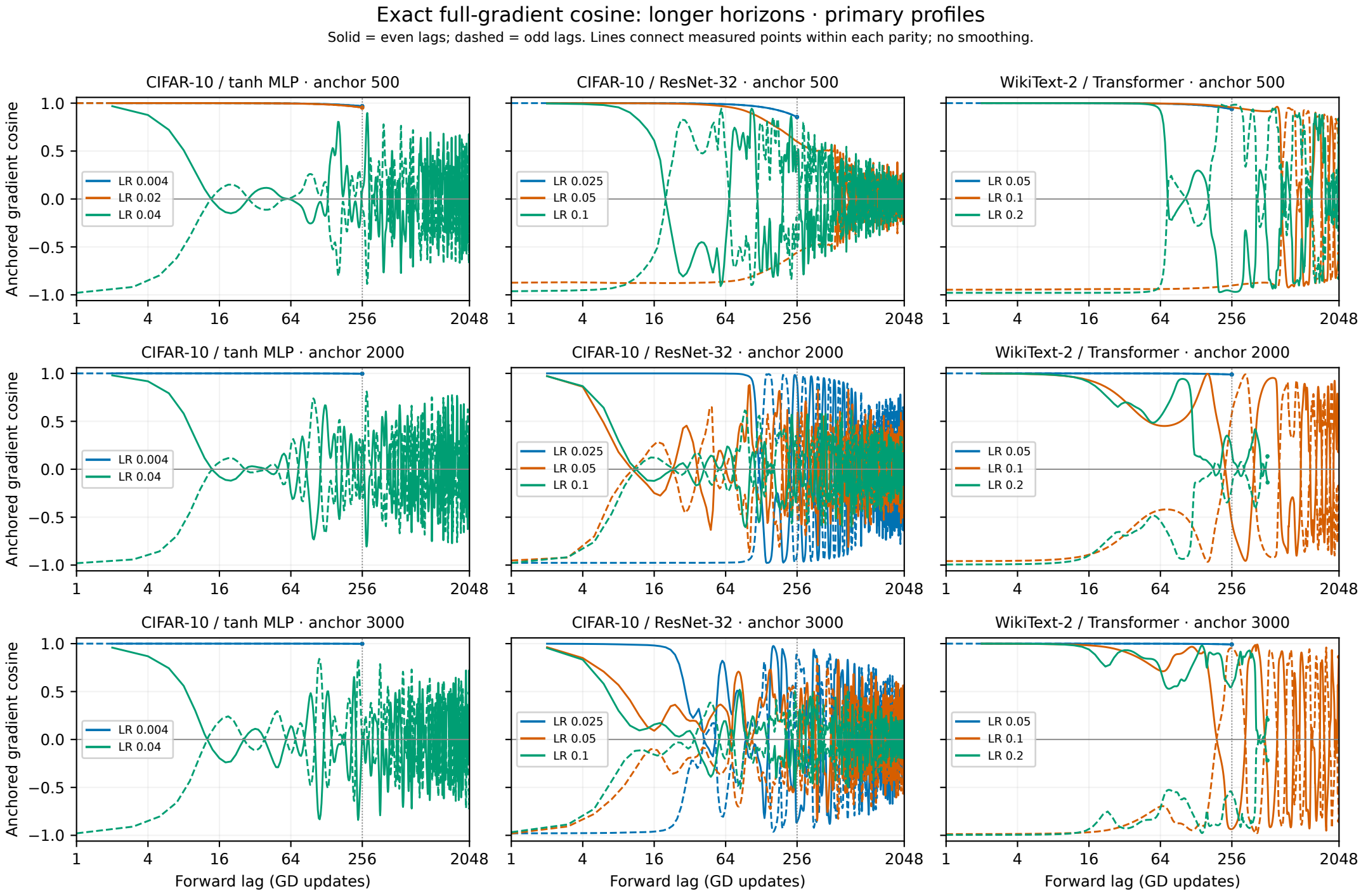

Exact full-gradient cosine: longer horizons · primary profiles
Solid = even lags; dashed = odd lags. Lines connect measured points within each parity; no smoothing.
CIFAR-10 / tanh MLP · anchor 500
CIFAR-10 / ResNet-32 · anchor 500
WikiText-2 / Transformer · anchor 500
CIFAR-10 / tanh MLP · anchor 2000
CIFAR-10 / ResNet-32 · anchor 2000
WikiText-2 / Transformer · anchor 2000
CIFAR-10 / tanh MLP · anchor 3000
CIFAR-10 / ResNet-32 · anchor 3000
WikiText-2 / Transformer · anchor 3000
Anchored gradient cosine
Forward lag (GD updates)
LR 0.004
LR 0.02
LR 0.04
LR 0.025
LR 0.05
LR 0.1
LR 0.2
1.0
0.5
0.0
−0.5
−1.0
1
4
16
64
256
2048
One anchored pair per lag, seed 0. Each curve ends at its declared adaptive stop; dotted vertical line = common lag 256.